

MAIN CODE

```
module mux_2to1 (  
    input a, b,  
    input sel,  
    output y );  
    assign y = (sel) ? b : a;  
endmodule
```

TEST BENCH CODE

```
module mux_2to1_tb();  
    reg a, b, sel;  
    wire y;  
    mux_2to1 uut (.a(a), .b(b), .sel(sel), .y(y));  
    initial  
    begin  
        $monitor("a=%b b=%b sel=%b y=%b", a, b, sel, y);  
  
        a=0; b=0; sel=0; #10;  
        a=0; b=1; sel=0; #10;  
        a=1; b=0; sel=0; #10;  
        a=1; b=1; sel=0; #10;  
  
        a=0; b=0; sel=1; #10;  
        a=0; b=1; sel=1; #10;  
        a=1; b=0; sel=1; #10;  
        a=1; b=1; sel=1; #10;  
  
        $finish;  
    end  
endmodule
```

MUX 2:1 WAVEFORM



MUX 2:1 DIAGRAM

